

Quiz on Matching Model of the Labor Market

Pascal Michailat

Question 1

In the matching model, when we derive the labor supply, we assume that:

- A) Inflows into unemployment equal outflows from unemployment.
- B) Inflows into unemployment are larger than outflows from unemployment.
- C) Inflows into unemployment are smaller than outflows from unemployment.
- D) Inflows into unemployment equal inflows into the labor force.
- E) Inflows into employment equal inflows into the labor force.
- F) None of the above.

Question 2

Consider a matching model of the labor market with labor force of size H , a recruiting cost of $r > 0$ recruiters per vacancy, a job-separation rate $s > 0$, and a Cobb-Douglas matching function: $m = \omega \times U^\eta \times V^{1-\eta}$. We define labor market tightness as $\theta = V/U$. Compute labor supply L^s .

- A) $L^s(\theta) = \frac{f(\theta)}{s \times f(\theta)} \times H$ where $f(\theta) = \omega \times \theta^{1-\eta}$
- B) $L^s(\theta) = \frac{f(\theta)}{s+f(\theta)} \times H$ where $f(\theta) = \omega \times \theta^{-\eta}$
- C) $L^s(\theta) = \frac{f(\theta)}{s+f(\theta)} \times H$ where $f(\theta) = \omega \times \theta^{1-\eta}$
- D) $L^s(\theta) = f(\theta) \times H$ where $f(\theta) = \omega \times \theta^{1-\eta}$
- E) $L^s(\theta) = \frac{s}{s+f(\theta)} \times H$ where $f(\theta) = \omega \times \theta^{1-\eta}$
- F) None of the above

Question 3

The labor supply $L^s(\theta)$ from the previous question has the following properties:

- A) It is increasing and concave in θ with $L^s(0) = 0$ and $L^s(\infty) = H$.
- B) It is increasing and convex in θ with $L^s(0) = 0$ and $L^s(\infty) = H$.

- C) It is decreasing and concave in θ with $L^s(0) = H$ and $L^s(\infty) = 0$.
- D) It is decreasing and convex in θ with $L^s(0) = H$ and $L^s(\infty) = 0$.
- E) It is increasing and concave in θ with $L^s(0) = 0$ and $L^s(\infty) = \infty$.
- F) It is increasing and convex in θ with $L^s(0) = 0$ and $L^s(\infty) = \infty$.
- G) None of the above.

Question 4

Why is the labor supply increasing in labor market tightness in the matching model?

- A) A higher tightness makes it more expensive to hire producers.
- B) A higher tightness makes it cheaper to hire producers.
- C) A higher tightness makes it easier to fill vacancies.
- D) A higher tightness makes it easier to find jobs.
- E) A higher tightness reduces the job-separation rate.
- F) None of the above.

Question 5

If the labor-force participation rate suddenly increases, what necessarily happens in the matching-model diagram?

- A) The labor supply curve is not affected.
- B) The matching function is more effective.
- C) The matching functions is less effective.
- D) The labor supply curve shifts inward.
- E) The labor supply curve shifts outward.
- F) None of the above.

Question 6

In the matching-model diagram,, what would an increase in the job-separation rate do?

- A) It would have no effect on the labor supply curve.
- B) It would shift the labor supply curve inward.
- C) It would shift the labor supply curve outward.
- D) It would make the matching function more effective.
- E) It would make the matching function less effective.
- F) None of the above.

Question 7

Consider a matching model of unemployment with labor force of size H , a recruiting cost of $r > 0$ recruiters per vacancy, a job-separation rate $s > 0$, and a Cobb-Douglas matching function: $m = \sqrt{U} \times \sqrt{V}$. Define labor market tightness as $\theta = V/U$. Using the assumption that labor-market flows are balanced, compute the recruiter-producer ratio $\tau = R/N$.

- A) $\tau(\theta) = \frac{\sqrt{\theta}}{1-r \times s \times \sqrt{\theta}}$
- B) $\tau(\theta) = \frac{r \times s}{1-r \times s \times \sqrt{\theta}}$
- C) $\tau(\theta) = \frac{r \times s \times \sqrt{\theta}}{1-r \times s \times \sqrt{\theta}}$
- D) $\tau(\theta) = \frac{r+s}{r+s \times \sqrt{\theta}}$
- E) $\tau(\theta) = \frac{r \times s \times \sqrt{\theta}}{r \times s \times \sqrt{\theta}-1}$
- F) None of the above

Question 8

The recruiter-producer ratio derived above has the following properties:

- A) It is increasing in θ and positive on $\mathbb{R}_+$, with $\lim_{\theta \rightarrow \infty} \tau(\theta) = \infty$.

- B) It is decreasing in θ and positive on $\mathbb{R}_+$, with $\lim_{\theta \rightarrow \infty} \tau(\theta) = 0$.
- C) It is increasing in θ and positive on $[0, rs]$, with $\lim_{\theta \rightarrow rs} \tau(\theta) = \infty$.
- D) It is increasing in θ and positive on $[0, 1/rs]$, with $\lim_{\theta \rightarrow 1/rs} \tau(\theta) = \infty$.
- E) It is decreasing in θ and positive on $[0, rs]$, with $\lim_{\theta \rightarrow rs} \tau(\theta) = 0$.
- F) None of the above.

Question 9

Consider a matching model of unemployment with labor force H , a recruiting cost of $r > 0$ recruiters per vacancy, a job-separation rate $s > 0$, a Cobb-Douglas matching function $m = \sqrt{U} \times \sqrt{V}$, a fixed wage w , and a production function $y = 2 \times a \times \sqrt{N}$, where a governs labor productivity and N denotes the number of producers in the firm. Define labor market tightness as $\theta = V/U$. What is the labor demand?

- A) $L^d(\theta) = (1 - rs\sqrt{\theta})^2 \times (a/w)^2$
- B) $L^d(\theta) = \frac{(w/a)^2}{(1 - rs\sqrt{\theta})^2}$
- C) $L^d(\theta) = \frac{(a/w)^2}{1 - rs\sqrt{\theta}}$
- D) $L^d(\theta) = (1 - rs\sqrt{\theta}) \times (a/w)^2$
- E) $L^d(\theta) = (1 - rs\sqrt{\theta}) \times (a/w)$
- F) None of the above

Question 10

The labor demand curve derived in the previous question has the following properties:

- A) It is decreasing in θ , with $L^d(0) = (a/w)^2$ and $L^d(1/(rs)^2) = 0$.
- B) It is decreasing in θ , with $L^d(0) = \infty$ and $L^d(\infty) = 0$.
- C) It is increasing in θ , with $L^d(0) = 0$ and $L^d(1/(rs)^2) = (a/w)^2$.
- D) It is decreasing in θ , with $L^d(0) = (a/w)$ and $L^d(1/(rs)) = 0$.
- E) None of the above.

Question 11

Consider a matching model with fixed wage. Imagine that the government implements training programs to increase the skills and productivity of workers. In the tightness-employment diagram, this policy would

- A) Shift the labor demand curve upward
- B) Shift the labor demand curve downward
- C) Shift the labor supply curve leftward
- D) Shift the labor supply curve rightward
- E) Rotate the labor demand curve upward
- F) Rotate the labor demand curve downward
- G) Have no effect on labor demand and labor supply
- H) None of the above

Question 12

In the matching model, which of the following parameters and variables negatively influence labor demand?

- A) Labor market tightness and productivity
- B) Wage and productivity
- C) Wage and labor market tightness
- D) Labor force and wage
- E) Labor force and recruiting cost
- F) None of the above